Preserve meaning at the factory handover

Factory Relay: on-device learning for multilingual factory communication

REPORT

The operator uses their preferred language or dialect.

INTERPRET

Local support translates into the recipient's chosen variety.

CONFIRM

Both agree on the asset, action and release conditions.

Research question: does reviewed local adaptation improve operational understanding?

HSE: industrial handover and cross-checking [1]

A real motivation; an expanded research plan

The available DSV account concerns warehouse support, not a verified manufacturing incident.

BUILT TODAY

Animated factory scenarios
Two complementary roles
Local typed-message play
Deterministic debrief

PLANNED EXTENSION

Local automated translation
Cross-device multiplayer
Optional personal training
Guarded laboratory control

First review: validate one company workflow and one community-reviewed language pair or variety.

Download once; interpret on the device

Browser game + local companion + versioned model pack + private data vault

CAPTURE

Original text or audio
Selected source profile
Separate ASR draft

MEDIATE

Approved local examples
Small language model
Recipient target variety

REVIEW

Original + translation
Uncertainty + check-back
Reviewed local records

Laptop first. Verify model compatibility and resources; reuse the cache. No silent cloud fallback.

Candidate text model: Qwen3-1.7B [3]

Candidate speech model: Qwen3-ASR-0.6B-hf [4]

After every session: review, then choose

Keep or delete → review examples → update memory and/or train → test → activate or roll back

MEMORY UPDATE

Store approved terms and examples for retrieval.
No model weights change.

PERSONAL ADAPTER

Freeze the base model; train a small LoRA adapter.
Model behavior can change.

Train only on reviewed examples. Keep a held-out check and activate adapters between sessions.

Too little verified data or device capacity? Queue examples and keep the current model.

PEFT: quantization and adapter training [5]

Play across devices; keep training data local

Encrypted peer channels exchange the selected chat and game events.

ON EACH DEVICE

Original recordings
Personal reviewed examples
Local memory + adapter
Encrypted data vault

SHARED FOR PLAY

Chosen messages
Room and role events
Confirmed game actions
Recipient-visible context

WebRTC needs connection infrastructure. Recipients can retain shared messages; privacy is not absolute.

WebRTC: peer connections, signaling and connectivity [6]

A guarded bridge to laboratory equipment

Hardware-dependent milestone: a small supervised conveyor or PLC demonstration

PROPOSE	Language model produces an explicit action proposal.
CONFIRM	A human verifies asset, action and release conditions.
AUTHORIZE	Gateway checks allowed commands, fresh state and interlocks.
EXECUTE	The controller performs the permitted action and returns state.

Stopping and network-loss handling remain independent of the language model.

OPC UA can provide an interface; it is not a safety guarantee [10]

Evaluate preserved meaning and real learning

Same base model: frozen → local memory → memory + personal adapter

PRIMARY OUTCOMES

Correct factory decisions
Asset / action / negation
Quantity / time / condition
Unsupported additions

SUPPORTING MEASURES

Clarification and latency
Memory / energy / time
Dialect-faithful WER/CER
Explicitly defined SAER

Matched, held-out scenarios; reviewed meanings; counterbalanced order. Report failures as well as gains.

ASR disparities are documented [2]

SAER definitions must be specified [8]

Deliver a local learning system in stages

Each milestone produces something demonstrable, with its remaining limits stated.

WEEKS 1–4

Company problem sheet
Language + device pilot
Reviewed scenarios
Proposal and demo

WEEKS 5–12

Local model delivery
Translation + remote play
Review + memory tools
Adapter training + rollback

WEEKS 13–15

Controlled evaluation
Guarded lab demonstration*
Failure analysis
Report and presentation

Risks: device capacity, target-language coverage, overfitting, company access and hardware availability.

* Hardware-dependent. Factory supervisor: process realism. AI/language supervisor: models and evaluation.